Team 6 – Element B

B1: Creating safe and enriching environments for turtles in captivity has been a major focus of research and animal care guidelines. These efforts address the challenges of trying to replicate natural habitats, managing their health risk and supporting their natural behaviors. One of the key guidelines that should be followed when thinking about habitats for animals in captivity is from the Institutional Animal Care and Use Committee (IACUC). These protocols outline how to properly care for animals in captivity which include the dimensions of the tank, as well as their water quality, and their basking areas and lighting. Rehabilitation centers and zoos as well as Towson University follow these rules to make sure that ethical and effective practices are taken place. For example tanks have to be large enough for the species and size of the animal, allowing them to move freely and act naturally. Water has to be kept in clean and with proper filtration, temperature control and salinity. Turtles also require a dry basking platforms to regulate their body temperature, and proper UVB lighting to support their shell health and calcium metabolism. These are crucial standards that protect these species and ensures that the rehabilitation efforts are ethical and effective.

The first peer reviewed article, "Rehabilitation of Marine Turtles and Welfare Improvement by Application of Environmental Enrichment Strategies," by Escobedo Bonilla et al. (2022), focus their study on enrichment strategies used in marine turtles in a rehabilitation center in Latin America. The results of these experiments are applicable globally. The study highlights that adding things like floating toys, different textures on the tank floor, and realistic tank set ups helps turtles act more like they would when they are in the wild. These natural behaviors are important for keeping turtles healthy in captivity.

Another study, "A Review of Welfare Indicators for Sea Turtles Undergoing Rehabilitation, with Emphasis on Environmental Enrichment" by Diggins et al. (2022), identifies the indicators of welfare with activity levels, their feeding behaviors , and stress responses. This could be a great thing to think about when building habitats and making sure the turtles are actually health and overall good. It emphasizes the importance of UVB lighting, basking platforms and clean water systems to prevent diseases like shell rot and metabolic bone disease.

The third article, "Captive Rearing Duration May be More Important than Environmental Enrichment for Enhancing Turtle Head Starting Success" by Tetzlaff et al. (2019), which does compare the effects of enrichment and rearing duration on success of head-started turtles. This study showed that keeping turtles in captivity for longer periods of time helped them survive better after being relieved, but giving them things like places to hide, climb, or explore also helped them grow properly and act more like they would in the wild.

Lastly, the study, "Aging Dynamics in Captive Sea Turtles Reflect Conserved Life-History Patterns Across the Testudines Phylogeny" by Glen et al. (2025) investigates how turtles age in captivity and compare to those in the wild. Researchers discovered that turtles living in captivity age in the same way as turtles in the wild. Because of this, it is important to keep their environment stable, like making sure the temperature is right, giving them proper UVB light, and feeding them a healthy diet so that they stay healthy over time.

These sources provide a strong foundation for understanding turtles habitats in captivity. They highlight what has been successful and what still needs improvement in creating environments that truly support the health and wellbeing of turtles.

B2: Each of these studies provides valuable insight into how to design a manage a turtle habitat a lot more effectively. One of the major success has been the use of environmental enrichment to encourage natural behavior and reduce stress. For example, Escobedo-Bonilla et al found that putting things like toys and different textures on the tank floor helped turtles move around more and feel less stressed, which made their behavior in captivity healthier. Diggins and their team found clear ways to check the well-being of a turtle like watching how active it is and how well it eats so caretakers can predict their outcomes based on the turtles conditions. Glen et al found that turtles in captivity still follow natural aging patterns, which reinforces the importance of mimicking their habitats temperature and lighting. And according to Tetzlaff et al, keeping turtles in captivity for a longer time before releasing them can help them survive better but only if their living conditions are well taken care of.

However, there are several challenges that these studies don't fully address. One major issue is the difficulty of cleaning and draining large tanks which something that can affect habitat maintenance. Many enrichment focused enclosures also make medical testing or close observation very difficult, which can delay health checks. Additionally, while improper UVB exposure can lead to metabolic bone disease, few studies offer clear solutions on how to balance providing enough warmth without over exposing. Overall these studies represent strong and credible prior research. They highlight both what works as well as how it still needs improvement, giving good ideas of how clear pictures of how to create a health and sustainable environment for these turtles in activity.

References:

- I have used AI for brainstorming research as well as helping me write this.
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